Full Teaching Notes

Session 3 — Full Teaching Notes (comprehensive per-slide guide)

The comprehensive per-slide instructor guide — English, LTR

Course context

- **Session 3 of 32.** 60 min teaching + 60 min problem-solving (separate meeting).
- **Goal (student language):** "By the end of today, YOU will be able to explain why one question is better than another."
- **Before class — verify all of this:**
  - notebook `s03_notebook.ipynb` runs top-to-bottom (PART 0 demo cell is the hook's machine beat)
  - three grab bags prepared: Bag 1 = 8 blue balls; Bag 2 = 4 blue + 4 red; Bag 3 = 3 blue + 3 red + 2 green (or colored magnets + two drawn boxes)
  - slide assets exist in `assets/`
  - the judge passes on the solution notebook (3/3)
- **Session timing map:** hook 0–5 · sortedness 5–15 · surprise 15–25 · the ranking game 25–35 · Titanic 35–45 · you-be-the-machine 45–50 · quiz 50–55 · handout 55–60.
- **Board plan:** the hook's two boxes stay drawn ALL session (Segment A names purity on them); vocabulary column on the right (purity, entropy, information gain — never erased); the grab-bag ratings (0, 1, 1.56) stay visible; the score card from Segment C stays up so Segment D's Titanic leaderboard lands under it, same shape.
- **The session's big move:** S2 ended on a mystery (WHO chose `weight ≥ 14`?). Today the class plays the machine's game by hand, then watches the machine play it — and the mystery dissolves into arithmetic.

Slide 1 — Title: درخت چطور سؤالش را انتخاب می‌کند؟ (lead)

Welcome them back as a show that keeps revealing its tricks.

Say: "Last time you drew trees by hand and the machine's tree asked `weight ≥ 14`. Today you find out where that question CAME from — and by the end, you'll be the one who can explain it."

Do: notebook open at PART 0; grab bags within reach but hidden; board clean except the top half (the hook needs it).

Slide 2 — دو جعبه، یک سؤال (hook, beat 1)

[0:00] The two boxes. Draw (or place) Box A: 4 blue + 4 red mixed. Box B: 8 blue.

Say: "The goal of today's whole game: get every box to be ONE color. Quick — which box needs help, and which one is already done?"

Expect: instant answers. "Box B is done" — that intuition IS purity; you are 40 minutes from naming it.

Do NOT say "sorted" or "pure" yet. Experience first, name later.

Slide 3 — کدام سؤال بهتر مرتب می‌کند؟ (hook, beat 2)

The vote. A new box: 2 blue + 2 red (mixed big/small). Three candidate questions on the slide: "is it red?", "is it big?", "is it round?". Hands up for each.

Say: "One question. Two doors. Which question gets this box to one color per side?"

Expect: "is it red?" wins obviously — good. Some will ask why we're voting on something so easy. Answer: "Because the machine has to vote on MILLIONS of these — and it can't go by gut."

Keep the votes on the board. The losing questions return on slide 6.

Slide 4 — ماشین به سؤال‌هایش نمره می‌دهد (hook, beat 3)

[RUN: PART 0 demo cell] — the leaderboard: every candidate question on the 11 animals with its score; `weight ≥ 14` on top.

Say: "Last session I promised you'd learn why 14. Here it is: the machine SHOPS. It tries every question on the menu, scores each one, and the champion gets asked. Same game you just played — with a number on it."

Expect: "It tried ALL the questions?!" — yes. The S2 mystery is answered in front of them.

Fallback if the cell fails: slide 13 carries the leaderboard picture — narrate from it. Never skip the story.

Slide 5 — خلوص (Purity) (definition)

[5:00] Segment A. NAME slide — the experience (two boxes + the vote) already happened.

Do: point at Box A and Box B on the board while reading: the good question leaves each side as one thing; the useless question cuts without sorting.

Write on the board (vocabulary column, never erased): خلوص (Purity) = how close a box is to ONE thing per room.

[MISCONCEPTION]: "a good question splits the data evenly" — NO. Splitting evenly is easy and worthless; SORTING is the goal. A question can halve the box and leave both sides as mixed as before (slide 6 shows exactly this).

CHECK (cold call): "The elephant's side of our S2 tree — the `weight ≥ 14 → yes` corridor — was it mixed or pure?" (Pure — all big dogs. THAT's why the tree stopped asking after one question. S2's asymmetry mystery gets its answer right here.)

Slide 6 — سؤالِ بیهوده (wow)

SHOW the useless question: `ear ≥ 4.5` → left: 1 cat · right: 10 animals, all three groups.

Say: "The box got CUT — one animal on one side, ten on the other. Did anything get SORTED? Look at the balls: both sides are exactly as mixed as before. We drew a line inside the mess."

Point back at the slide-3 votes: "the 'is it round?' votes — this is them, in the machine's world."

Slide 7 — سؤالِ قهرمان (wow)

SHOW the champion: `weight ≥ 14` → left: 7 animals (4 cats + 3 small dogs) · right: 4 big dogs, PURE.

Say: "Same box, same two doors — but THIS question sorted. The yes-side is one thing: big dogs, pure, done. The no-side is still mixed — and that's exactly why the tree asks a SECOND question there (slide 14: you will pick it yourselves)."

Do: trace the champion with your finger, door by door. Let the difference between slides 6 and 7 sink in — this difference is what the machine measures.

Slide 8 — مرتب شد یا نه؟ (quiz)

USE-IT (2 min). Before commenting: the class votes on `weight ≥ 30` (no: 9 mixed animals · yes: 2 big dogs) — sorted, or just cut?

Answer: PARTIALLY sorted — the yes-side is pure (2 big dogs), the no-side stays mixed. A real question, just not the champion: its gain is 0.32 against 0.95.

Why this slide exists: "partially sorted" is the honest middle — most real questions live there. Purity is a scale, not a yes/no.

Slide 9 — کیسه‌های غافلگیری (hook — the grab-bag game)

[15:00] Segment B. THE GAME (3 min, high energy). Three paper bags: 8 blue · 4 blue + 4 red · 3 blue + 3 red + 2 green. A volunteer draws ONE ball blind from each. Before each draw, the class SHOUTS its prediction. After each draw, everyone rates their surprise with fingers: 0, 1, or 2.

Expect: Bag 1 = zero surprise (you know it's blue). Bag 3 = maximum surprise. The class just invented the ranking: pure = 0, mixed = big.

Energy note: play it fast and loud. This beat carries the whole concept.

Slide 10 — غافلگیری (آنتروپی) (definition)

NAME slide. Write the three bags' numbers on the board: 0, 1, 1.56. (From counting colors — no formula today.)

Say: "That feeling you rated with fingers — the machine needs it as a NUMBER. Here it is: pure box = 0, the even red/blue mix = 1, the three-color bag = 1.56. The messier the box, the bigger the number."

[MISCONCEPTION]: entropy is NOT the machine thinking. It is a number from counting colors. Nothing thinks anywhere — this is the LEAST magical part of the whole course, and saying so plainly increases trust.

[RUN: cell A2] — the machine prints the same three numbers next to the color counts. Check them together against the finger-ratings: same ranking.

Slide 11 — جعبهٔ ۱۱ حیوان: چند؟ (quiz)

USE-IT (cell A2b). The 11-animal box: 4 cats, 4 big dogs, 3 small dogs. The class predicts the surprise level — "middle?" — then runs.

Answer: 1.57 — nearly the three-color bag. Same kind of mix: three things, none dominating.

Say: "Every box has its number now. And a question's quality is just... how much it LOWERS that number. Which brings us to the game."

Slide 12 — بازی: تو امتیاز بده (hook — EXPERIENCE first)

[25:00] Segment C. THE GAME (5 min, teams of two). No vocabulary on this slide — the class plays first.

Candidate questions on the 11 animals (in `content.md` Segment C): `weight ≥ 14`, `weight ≥ 3.25`, `ear ≥ 8.25`, `ear ≥ 4.5` — each with its two doors written out on the board.

Teams compute the score card (mess before - mess after); first team with the right champion shouts.

Result: a TIE — `weight ≥ 14` and `ear ≥ 8.25` both drop 1.57 → 0.63 (gain 0.95). The "even splits" barely move the surprise (1.43 after).

Why the tie is gold: both champions do the same amount of sorting. The machine breaks the tie mechanically — first column measured wins (weight). Honest and boring — say so.

Slide 13 — نمرهٔ سؤال: بهرهٔ اطلاعاتی (definition — NAME after play)

[~32:00] NAME slide — AFTER the game, never before. What the teams just computed by hand is called بهرهٔ اطلاعاتی (Information Gain): غافلگیریِ قبل منهای غافلگیریِ بعد — weighted by how many animals land on each side. Champion = biggest gain.

[MISCONCEPTION]: the champion is NOT the purest side — it is the biggest surprise-drop for the WHOLE box. Point BACK to the bad question (slide 6): its «no» door was perfectly pure yet scored only 0.14, because it held one animal. Corrected sentence, with the class: "a good question makes BIG groups purer."

Slide 14 — جدول امتیازها (wow)

[RUN: cell A3] — the machine's leaderboard, sorted, champion on top — the same ranking the class computed by hand.

Say: "You played its game. It plays this game on EVERY node of EVERY tree — millions of times per tree. There is no committee, no committee meeting. Just this table."

Point at the tie: "Both champions gain 0.95. The machine keeps the first one it measured — weight is column one. Boring. Honest. Arithmetic."

Slide 15 — راند دوم: جعبهٔ سبک (quiz)

USE-IT (cell A3b). The light box: 7 animals (4 cats + 3 small dogs), surprise 0.99. The class predicts the champion BEFORE running.

Answer: `ear ≥ 8.25` — takes the box to ZERO surprise (4 cats | 3 small dogs, both sides pure).

The payoff moment: "That is the SECOND question of Session 2's tree. You just derived it yourselves — the machine's tree was never magic; it was this game, played twice."

Slide 16 — ماشین این بازی را روی تایتانیک هم می‌کند (hook)

[35:00] Segment D — THE MOMENT. Students code along cells A4–A5 (typing, not watching).

[RUN: cell A4] — the Titanic box: 712 passengers, 288 survived, 424 did not. Surprise: 0.97.

Say: "Session 1's data, back on screen. A very mixed box — barely better than a coin flip. If you were the machine, where would you start sorting?"

Slide 17 — قهرمانِ تایتانیک (wow)

[RUN: cell A5] — the leaderboard: `is_female` 0.214 · `pclass` 0.083 · `fare` 0.069 · `age` 0.018.

SHOW the before/after boxes: men (453 passengers, 21% survived) and women (259 passengers, 75% survived) — both far more one-color than the full box.

THE CALLBACK (say it warmly): "Session 1, I had you VOTE on which feature matters for survival. Most of you said gender. You beat the machine to it — it just put a NUMBER on your vote: 0.21, triple the runner-up."

[MISCONCEPTION] (one sentence only): the gain number means gender SORTS the passengers — it does not mean gender CAUSED survival. Causes are a different door (Session 9 opens it properly). Do not open it now.

Slide 18 — کلِ راز (codespot)

[45:00] Segment E — you be the machine.

[RUN: cell A6] — the whole secret in plain Python: loop over every question, compute the gain, take the max. Then the same call on the light box. The S2 tree's two questions come out.

REBUILD the S2 tree on the board, as a class: root box → champion (`weight ≥ 14`) → light box still mixed → shop again → `ear ≥ 8.25`. Ninety seconds, door by door, from nothing but the score card.

HONEST BEAT (say it plainly): "The machine doesn't think. It measures and takes the max — like a scale weighing fruit. When two questions tie, it keeps the first one it measured. No cleverness anywhere. And still — this is the engine inside every tree you'll ever train. THAT is the surprise: how far simple arithmetic goes."

Slide 19 — انتخابِ سؤال، کارِ ماست (dark)

The C2 rabbit returns. In Session 2's problem hour, no amount of depth or extra examples fixed the rabbit — because the fix was a question about something never measured.

Say: "The machine only shops among the table's columns. 'Does it hop?' is never asked — it isn't a column. WE choose what goes on the menu; the machine picks from it. Feature choice is a human job — and when we feed garbage, the champion of garbage is still garbage. (That story has its own session: Session 7.)"

Energy note: this slide is deliberately slow and serious — one minute, then back up for the quiz.

Slide 20 — امروز یاد گرفتی (quiz)

[50:00] CONSOLIDATION — cold call, one question per student:

1. What does a "good" question do? (raises purity — leaves each side mostly one thing)
2. What is entropy, in kid words? (the surprise level of a box — 0 when pure)
3. How does the machine score a question? (surprise before - after; biggest drop wins)
4. Why did the machine pick `weight ≥ 14` for the animals? (biggest gain: 1.57 → 0.63)
5. What won on Titanic? (is_female — the class's own S1 vote, with a number on it)

If anyone stumbles on Q1–3, re-explain NOW, not at home.

Slide 21 — تحویل امشب + تمرین‌های خانه (handout)

[55:00] THE SHIP + at-home handout in one slide. The Ship is the annotated Titanic split plot — cell B3's leaderboard + before/after boxes, in their own hand, committed tonight. The slide shows the Ship art AND the live judge command.

Explain the loop OUT LOUD: 3 problems tonight (20–30 min); the next session opens by solving them TOGETHER from their attempts. First judge run is all-red — that is normal; say it.

The judge command is visible on the slide: `python judge/run.py s03_notebook.ipynb`

One sentence of story per problem:

- **B1** — the three bags get their numbers (0, 1, 0.47)
- **B2** — derive the light box champion (you did it in class — now alone)
- **B3** — Titanic's leaderboard + the annotated plot (the Ship)

Deadline: before the problem-solving session.

Slide 22 — جلسةٌ بعد (lead, cliffhanger)

[60:00 — end on time.]

Cliffhanger (verbatim): "Today the machine picked its questions by measuring surprise — on the animals it has SEEN. But here is a trap: what if a question is only 'good' on the passengers it studied? A student who memorizes every exam answer looks brilliant — until the new exam. Tomorrow: you catch the machine fooling itself."

After class (48h): fill the retro note in the session README.